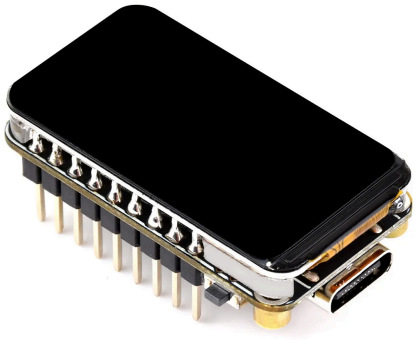
ESP32-C6-LCD-1.47

ESP32-C6-LCD-1.47 | ESP32-C6-LCD-1.47-M (With pin header)



NOTE

Do not remove the screen when soldering the pin headers. Tilt the soldering iron and solder the headers directly.

NOTE

Keep the screen brightness at 50% or lower during use, and do not operate the screen at full brightness for extended periods. Excessive brightness increases the screen temperature, and overheating may cause dark shadows on the screen and affect normal display.

If the display is already abnormal, allow the development board to cool down, then flash a program with a lower brightness setting.

Product Overview

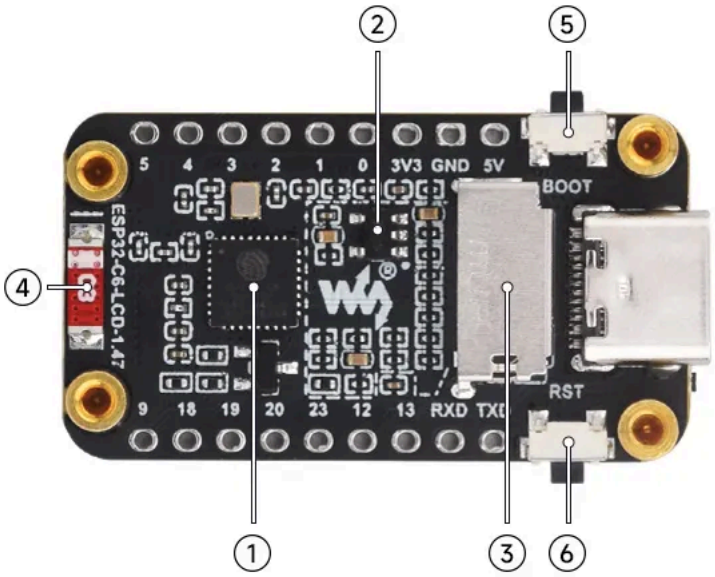
The ESP32-C6-LCD-1.47 is a microcontroller development board that supports 2.4 GHz Wi-Fi 6 and BLE 5. It integrates 4 MB Flash and has onboard 1.47-inch LCD screen, can smoothly run GUI programs such as LVGL. Combined with various peripheral interfaces, it is suitable for the quick development of the HMI and other ESP32-C6 applications.

SKU	Product
28563	ESP32-C6-LCD-1.47
30381	ESP32-C6-LCD-1.47-M

Features

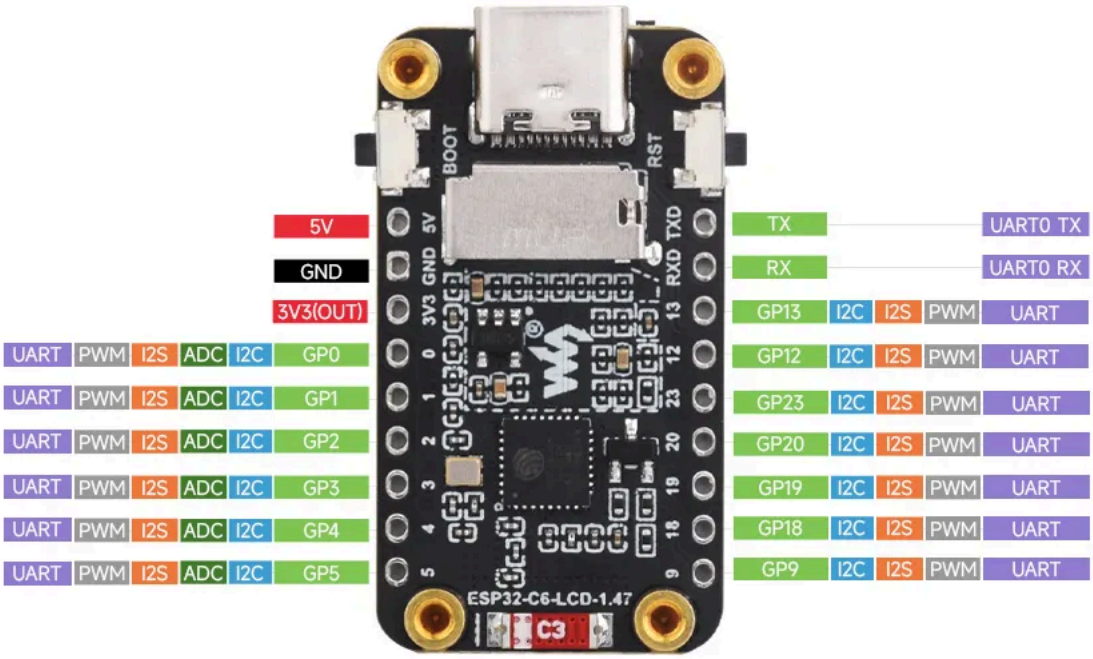
- Equipped with a high-performance RISC-V 32-bit processor and a low-power RISC-V 32-bit processor; the high-performance processor clock is up to 160 MHz and the low-power processor clock is up to 20 MHz
- Supports 2.4 GHz Wi-Fi 6 (802.11 ax/b/g/n) and Bluetooth 5 (BLE), with onboard antenna
- Onboard 320 KB ROM, 512 KB HP SRAM, 16 KB LP SRAM, and 4 MB flash
- Onboard 1.47-inch LCD screen, 172 × 320 resolution, 262K colors
- Most GPIOs are broken out, and full-speed USB serial port is integrated
- Onboard TF card slot for external TF card storage of images or files
- Supports flexible clock control and various power modes for low-power scenarios
- Onboard RGB color LED, which works together with the transparent acrylic interlayer to create cool lighting effects

Onboard Resources



1. **ESP32-C6FH4**
2. **ME6217C33M5G** Low dropout LDO, current (Max) 800 mA
3. **TF card slot**
4. **Onboard ceramic antenna**
5. **BOOT button**
6. **RESET Button**

Interface Introduction



- LCD

LCD Pin	ESP32C6
MOSI	GPIO6
SCLK	GPIO7
LCD_CS	GPIO14
LCD_DC	GPIO15
LCD_RST	GPIO21
LCD_BL	GPIO22

- RGB LED

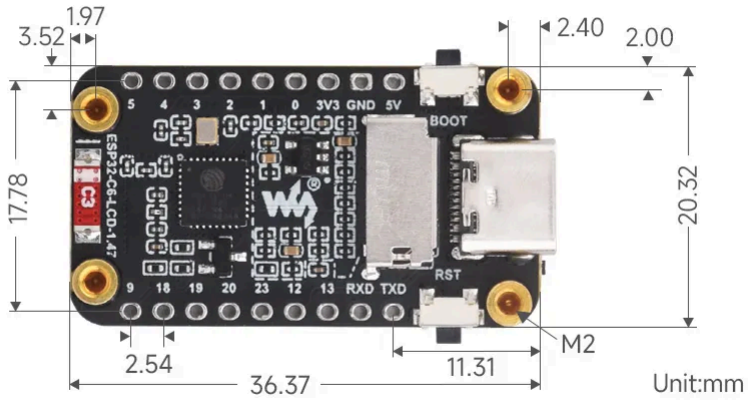
RGB LED	ESP32C6
RGB_Control	GPIO8

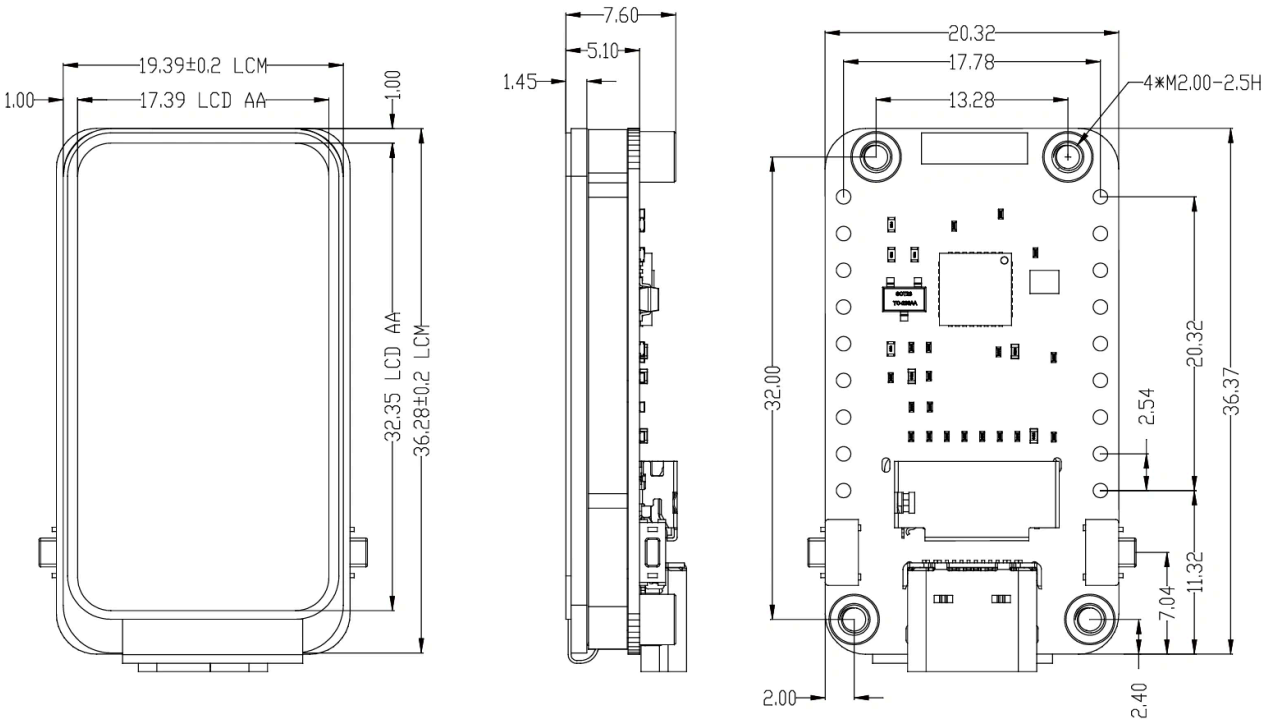
- TF Card

TF Card	ESP32C6
MISO	GPIO5

TF Card	ESP32C6
MOSI	GPIO6
SCLK	GPIO7
CS	GPIO4
SD_D1	NC
SD_D2	NC

Dimensions





Specifications

Parameter	Value
Interface	USB Type-C
Main Controller	ESP32-C6
Screen Type	TFT
Screen Controller	Display: ST7789
Onboard Device	Micro SD

Development Methods

> **NOTE**
>
> Keep the screen brightness at 50% or lower during use, and do not operate the screen at full brightness for extended periods. Excessive brightness increases the screen temperature, and overheating may cause dark shadows on the screen and affect normal display.
>
> If the display is already abnormal, allow the development board to cool down, then flash a program with a lower brightness setting.

The ESP32-C6-LCD-1.47 supports two development frameworks: Arduino IDE and ESP-IDF, offering flexibility for developers. You can choose the appropriate development tool based on project requirements and personal preference.

Arduino generally offers a gentler learning curve and may be more approachable for beginners and hobbyists. ESP-IDF provides advanced tooling and finer control over system behavior, making it better suited to complex projects and applications with demanding performance requirements.

- **Arduino IDE** is a convenient, flexible, and easy-to-use open-source electronics prototyping platform. It requires minimal foundational knowledge, allowing for rapid development after a short learning period. Arduino has a huge global user community, providing a vast amount of open-source code, project examples, and tutorials, as well as a rich library ecosystem that encapsulates complex functions, enabling developers to implement various features rapidly. You can refer to the **Working with Arduino** to complete the initial setup, and the tutorial also provides related example programs for reference.

- **ESP-IDF**, short for Espressif IoT Development Framework, is a professional development framework launched by Espressif Systems for its ESP series of chips. It is based on C language development and includes compilers, debuggers, flashing tools, etc. It supports development via command line or integrated development environments (such as Visual Studio Code with the Espressif IDF plugin), which provides features like code navigation, project management, and debugging. We recommend using VS Code for development. For the specific configuration process, please refer to the **Working with ESP-IDF**. The tutorial also provides relevant example programs for reference.

Give Feedback